

The full power growth of k -wise Khinchine constants

Literature-implied answer to the questions in arXiv:1412.7859

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Status: literature-implied full asymptotic answer. For fixed $k \geq 2$, $p \geq 2$, and $m = \lfloor k/2 \rfloor$,

$$C(N, p, k) \asymp_{p,k} N^{\max\{0, 1/2 - m/p\}}.$$

In particular, for every fixed $p > k$, the constants are unbounded and their growth rate is $N^{1/2 - \lfloor k/2 \rfloor / p}$.

Source questions

For a k -wise independent Rademacher vector $\varepsilon = (\varepsilon_i)_{i=1}^N$, the source defines

$$C(N, p, k) = \sup_{\substack{\|a\|_2=1 \\ \varepsilon \text{ } k\text{-wise independent}}} \left(\mathbb{E} \left| \sum_{i=1}^N a_i \varepsilon_i \right|^p \right)^{1/p}.$$

On source PDF page 2 it asks whether this is bounded for fixed p, k and, if not, for its growth rate. Page 3 says that understanding $k \geq 4$ remains open [1]. The revised paper restricts the numbered question to $p > k$ and proves the required general upper estimate for even k , while again leaving sharpness for $k \geq 4$ open [3].

Literature-implied theorem. Fix $k \geq 2$ and $p \geq 2$, and put $m = \lfloor k/2 \rfloor$. As $N \rightarrow \infty$,

$$C(N, p, k) \asymp_{p,k} N^{\max\{0, 1/2 - m/p\}}.$$

Hence $C(N, p, k)$ is bounded in N exactly when $p \leq 2m$.

Upper bound

Set $r = 2m \leq k$. Any k -wise independent Rademacher vector is r -wise independent. If $f = |\sum_i a_i \varepsilon_i|$, then expansion of the even power and r -wise independence give exactly the independent-sign moment:

$$\|f\|_r \leq C(r) \|a\|_2.$$

Also $\|f\|_\infty \leq \|a\|_1 \leq \sqrt{N} \|a\|_2$. Thus, for $p \geq r$,

$$\|f\|_p^p \leq \|f\|_\infty^{p-r} \|f\|_r^r, \quad \|f\|_p \leq C(r)^{r/p} N^{1/2 - r/(2p)} \|a\|_2.$$

This is Proposition 2.1 of [3], applied at the largest even order not exceeding k . If $2 \leq p \leq r$, monotonicity of L_q norms gives instead $\|f\|_p \leq \|f\|_r \leq C(r) \|a\|_2$. These are the claimed upper bounds in both regimes.

Lower bound from the maximal endpoint atom

Let $M(N, k, 1/2)$ be the largest possible probability of the all-ones vector under a k -wise independent law on N fair bits. Peled–Yadin–Yehudayoff prove, for every fixed k , the sharp asymptotic order

$$M(N, k, 1/2) \asymp_k N^{-\lfloor k/2 \rfloor} = N^{-m}; \quad (1)$$

see the odd- k reduction (1.1), Theorem 1.1, and the fixed- k conclusion after Corollary 1.3 on supporting PDF pages 4–7 [2].

Choose a law attaining the maximum (the feasible polytope is compact), write its bits as $X_i \in \{0, 1\}$, and set $\varepsilon_i = 2X_i - 1$. Then ε is a k -wise independent Rademacher vector. With $a_i = N^{-1/2}$, on the all-ones event

$$\sum_{i=1}^N a_i \varepsilon_i = \sqrt{N}.$$

Consequently, by (1),

$$C(N, p, k)^p \geq M(N, k, 1/2) N^{p/2} \text{gtrsim}_k N^{p/2-m},$$

and therefore

$$C(N, p, k) \gtrsim_{p,k} N^{1/2-m/p}.$$

This supplies the matching lower bound whenever $p > 2m$. When $p \leq 2m$, taking $a = e_1$ gives $C(N, p, k) \geq 1$, matching the bounded upper estimate. The theorem follows.

Proof intuition

Limited independence fixes every moment only up to the largest even order $2m \leq k$. Interpolating that controlled moment with the deterministic ceiling $\sqrt{N}$ produces the upper exponent. The complementary extremal law spends probability about N^{-m} at the most distant Hamming vertex. Equal coefficients turn this small atom into a value $\sqrt{N}$; its p th-moment contribution is exactly the missing power. The two estimates meet because both are governed by the same number m of controlled even moments.

Provenance and scope

The endpoint estimate (1) predates the source, but neither [2] nor the later Khinchine paper states the combined conclusion above. Conversely, the revised Pass–Spektor paper gives the matching upper bound but does not cite the endpoint estimate as a lower-bound mechanism. The result is therefore a direct implication of established literature, not a new construction.

The notation $\asymp_{p,k}$ means two-sided bounds by positive constants depending only on fixed p, k . This fully answers “bounded or not” and “growth rate” in the usual asymptotic-order sense; it does not determine an exact finite- N optimizer or an exact leading constant for $k \geq 4$.

References

- [1] B. Pass and S. Spektor, *On Khintchine type inequalities for pairwise independent Rademacher random variables*, arXiv:1412.7859v1 (2014), pp. 2–3.
- [2] R. Peled, A. Yadin, and A. Yehudayoff, *The Maximal Probability that k -wise Independent Bits are All 1*, arXiv:0801.0059v3 (2011), Theorem 1.1 and Corollary 1.3.
- [3] B. Pass and S. Spektor, *On Khintchine type inequalities for k -wise independent Rademacher random variables*, arXiv:1708.08775v1 (2017), Proposition 2.1.